



SYMBIOSIS INSTITUTE OF TECHNOLOGY, NAGPUR
Symbiosis International (Deemed University)

(Established under section 3 of the UGC Act, 1956)

Re-accredited by NAAC with 'A++' Grade | Awarded Category – I by UGC

Founder: Prof. Dr. S. B. Mujumdar, M. Sc., Ph. D. (Awarded Padma Bhushan and Padma Shri by President of India)

Experiment No.: 02

Title: Use var, let, const, and demonstrate basic features like template literals and destructuring; build a billing calculator with user input.

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>Billing Calculator</title>
```

```
</head>
```

```
<body>
```

```
<h2>Billing Calculator</h2>
```

```
<script>
```

```
const GST = 0.18;
```



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```
var customerName = prompt("Enter Customer Name: ");
```

```
var productName = prompt("Enter Product Name: ");
```

```
var quantity = Number(prompt("Enter quantity"));
```

```
let price = Number(prompt("Enter price per Item"));
```

```
let total = quantity * price;
```

```
let gstAmount = total * GST;
```

```
let finalBill = total + gstAmount;
```

```
const bill = {
```

```
    customer: customerName,
```

```
    product: productName,
```

```
    quantity,
```

```
    price,
```

```
    total,
```

```
    gstAmount,
```

```
    finalBill
```

```
};
```



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```
const {  
  
    customer,  
  
    product,  
  
    gstAmount: gst,  
  
    finalBill: amount  
}  
= bill;  
  
document.write(`<h3>Billing Receipt</h3>`);  
  
document.write(`<p>Customer: ${customer}</p>`);  
  
document.write(`<p>Product: ${product}</p>`);  
  
document.write(`<p>Quantity: ${quantity}</p>`);  
  
document.write(`<p>Price per Item: ₹${price}</p>`);  
  
document.write(`<p>Total: ₹${total}</p>`);  
  
document.write(`<p>GST (18%): ₹${gst.toFixed(2)}</p>`);  
  
document.write(`<h3>Final Bill: ₹${amount.toFixed(2)}</h3>`);  
  
</script>
```



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</body>

</html>

Output:

▼

📁

🌐 Billing Calculator

×

+

←

↺

📄 127.0.0.1:5500/practical_2.html

Billing Calculator

Billing Receipt

Customer: Aishwarya Brahmanekar

Product: Dell laptop

Quantity: 1

Price per Item: ₹24070521212

Total: ₹24070521212

GST (18%): ₹4332693818.16

Final Bill: ₹28403215030.16



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Case Study No.: 02

Title: To design and develop an interactive, client-side billing application (BILL.HTML) using HTML5, CSS3, and JavaScript (ES6+) that parses real-time product quantities, applies discount thresholds, calculates taxation rates, and renders an itemised invoice directly onto the Document Object Model (DOM).

Software/Tools Required: HTML5, CSS3, JavaScript, VS Code , Google Chrome

Experiment Program Code:

```
<!DOCTYPE html>
```

```
<html lang="en">
```

```
<head>
```

```
  <meta charset="UTF-8">
```

```
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
```

```
  <title>E-Commerce Automated Billing Engine</title>
```

```
  <style>
```

```
    body {
```

```
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
```

```
      line-height: 1.6;
```

```
      margin: 0;
```

```
      padding: 30px;
```

```
      background-color: #f7f9fc;
```



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```
}
```

```
.billing-container {
```

```
    max-width: 700px;
```

```
    background: #ffffff;
```

```
    margin: auto;
```

```
    padding: 25px;
```

```
    border-radius: 8px;
```

```
    box-shadow: 0 4px 15px rgba(0,0,0,0.05);
```

```
}
```

```
h2, h3 {
```

```
    color: #1a365d;
```

```
    text-align: center;
```

```
    margin-top: 0;
```

```
}
```

```
.student-badge {
```

```
    text-align: center;
```

```
    font-size: 14px;
```

```
    color: #718096;
```

```
    margin-bottom: 25px;
```



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border-bottom: 2px dashed #e2e8f0;

padding-bottom: 10px;

}

.form-row {

display: flex;

gap: 15px;

margin-bottom: 15px;

}

.form-group {

flex: 1;

display: flex;

flex-direction: column;

}

label {

font-weight: 600;

margin-bottom: 6px;

color: #2d3748;

}

input, select {



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```
padding: 10px;

border: 1px solid #cbd5e0;

border-radius: 5px;

font-size: 15px;

}

input:focus, select:focus {

    outline: none;

    border-color: #3182ce;

}

button {

    background: #2b6cb0;

    color: #ffffff;

    padding: 12px;

    border: none;

    border-radius: 5px;

    cursor: pointer;

    width: 100%;

    font-size: 16px;

    font-weight: bold;
```




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```
margin-top: 10px;

transition: background 0.2s;

}

button:hover {

    background: #2c5282;

}

.invoice-box {

    margin-top: 25px;

    padding: 20px;

    background: #f8fafc;

    border: 1px dashed #4299e1;

    border-radius: 6px;

    display: none;

}

.invoice-table {

    width: 100%;

    border-collapse: collapse;

    margin-top: 15px;

}
```



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```
.invoice-table th, .invoice-table td {
```

```
    padding: 10px;
```

```
    text-align: left;
```

```
    border-bottom: 1px solid #e2e8f0;
```

```
}
```

```
.invoice-table th {
```

```
    background: #edf2f7;
```

```
    color: #2d3748;
```

```
}
```

```
.text-right {
```

```
    text-align: right !important;
```

```
}
```

```
.total-highlight {
```

```
    font-size: 18px;
```

```
    font-weight: bold;
```

```
    color: #e53e3e;
```

```
}
```

```
</style>
```

```
</head>
```



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<body>

<div class="billing-container">

<h2>E-Commerce Billing System</h2>

<div class="student-badge">Developer: Aishwarya Brahmanekar | PRN: 24070521212</div>

<form id="billingForm" onsubmit="generateInvoice(event)">

<div class="form-row">

<div class="form-group">

<label for="customerName">Customer Name:</label>

<input type="text" id="customerName" placeholder="Enter name" required>

</div>

<div class="form-group">

<label for="invoiceNo">Invoice Number:</label>

<input type="text" id="invoiceNo" placeholder="INV-2026-001" required>

</div>

</div>

<div class="form-row">



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```
<div class="form-group" style="flex: 2;">
```

```
<label for="itemName">Item Description:</label>
```

```
<input type="text" id="itemName" placeholder="Product name/model" required>
```

```
</div>
```

```
<div class="form-group">
```

```
<label for="itemCategory">Category:</label>
```

```
<select id="itemCategory">
```

```
<option value="Electronics">Electronics (18% GST)</option>
```

```
<option value="Clothing">Clothing (5% GST)</option>
```

```
<option value="Essentials">Essentials (0% GST)</option>
```

```
</select>
```

```
</div>
```

```
</div>
```

```
<div class="form-row">
```

```
<div class="form-group">
```

```
<label for="unitPrice">Unit Price (₹):</label>
```

```
<input type="number" id="unitPrice" step="0.01" min="0" placeholder="0.00"
required>
```



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```
</div>

<div class="form-group">

  <label for="quantity">Quantity:</label>

  <input type="number" id="quantity" min="1" placeholder="1" required>

</div>

<div class="form-group">

  <label for="discountRate">Discount (%):</label>

  <input type="number" id="discountRate" min="0" max="100" placeholder="0">

</div>

</div>

<button type="submit">Compute Calculations & Generate Receipt</button>

</form>

<div id="invoiceBox" class="invoice-box">

  <h3>RETAIL INVOICE RECEIPT</h3>

  <p><strong>Customer Reference:</strong> <span id="invCustomer"></span><br>

  <strong>Invoice String ID:</strong> <span id="invId"></span></p>
```



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<table class="invoice-table">

<thead>

<tr>

<th>Item Description</th>

<th class="text-right">Rate</th>

<th class="text-right">Qty</th>

<th class="text-right">Subtotal</th>

</tr>

</thead>

<tbody>

<tr>

<td> ()</td>

<td class="text-right">₹</td>

<td class="text-right"></td>

<td class="text-right">₹</td>

</tr>

<tr>

<td colspan="3" class="text-right">Discount Applied:</td>

<td class="text-right" style="color: #38a169;">-₹</td>



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</tr>

<tr>

<td colspan="3" class="text-right">Calculated Tax Component (GST):</td>

<td class="text-right">₹</td>

</tr>

<tr style="background: #f7fafc;">

<td colspan="3" class="text-right total-highlight">Grand Total

Payable:</td>

<td class="text-right total-highlight">₹</td>

</tr>

</tbody>

</table>

</div>

</div>

<script>

function generateInvoice(event) {

event.preventDefault();



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// 1. Data Capture Extractors

```
var customer = document.getElementById("customerName").value.trim();

var invoiceNum = document.getElementById("invoiceNo").value.trim();

var item = document.getElementById("itemName").value.trim();

var category = document.getElementById("itemCategory").value;

var price = parseFloat(document.getElementById("unitPrice").value);

var qty = parseInt(document.getElementById("quantity").value);

var discRate = parseFloat(document.getElementById("discountRate").value) || 0;
```

// 2. Arithmetic Matrix Calculations

```
var rawSubtotal = price * qty;

var discountAmount = (discRate / 100) * rawSubtotal;

var discountedSubtotal = rawSubtotal - discountAmount;
```

// Tax Percentage Matrix Assignment

```
var taxPercentage = 0;

if (category === "Electronics") taxPercentage = 18;

else if (category === "Clothing") taxPercentage = 5;

else if (category === "Essentials") taxPercentage = 0;
```




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```
var taxAmount = (taxPercentage / 100) * discountedSubtotal;
```

```
var grandTotal = discountedSubtotal + taxAmount;
```

```
// 3. UI Node Rendering Updates
```

```
document.getElementById("invCustomer").innerText = customer;
```

```
document.getElementById("invId").innerText = invoiceNum;
```

```
document.getElementById("invItem").innerText = item;
```

```
document.getElementById("invCat").innerText = category;
```

```
document.getElementById("invPrice").innerText = price.toFixed(2);
```

```
document.getElementById("invQty").innerText = qty;
```

```
document.getElementById("invSub").innerText = rawSubtotal.toFixed(2);
```

```
document.getElementById("invDisc").innerText = discountAmount.toFixed(2);
```

```
document.getElementById("invTax").innerText = taxAmount.toFixed(2);
```

```
document.getElementById("invGrand").innerText = grandTotal.toFixed(2);
```

```
// Slide/Toggle element block presence configurations
```

```
document.getElementById("invoiceBox").style.display = "block";
```

```
}
```

